

HDOC-DAY6

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Batch no.- 16

Question 1:

Write a C program to find the area in square centimetres, perimeter in centimetres, and diagonal in centimetres of a rectangle when its length and breadth are given in meters.

Task1: Understand the Problem Statement:

- Our objective is to develop a C program that converts the length and breadth of a rectangle from meters to centimetres and calculates:
 - i) Area of the rectangle (in centimetres)
 - ii) Perimeter of the rectangle (in centimetres)
 - iii) Length of the diagonal (in centimetres)
- Required input for the C program will be:
 - Length of the rectangle in meters
 - Breadth of the rectangle in meters
- Formulae used in the program are as follows:
 - $\text{Length (cm)} = \text{Length (m)} \times 100$
 - $\text{Breadth (cm)} = \text{Breadth (m)} \times 100$
 - $\text{Area} = \text{Length} \times \text{Breadth}$
 - $\text{Perimeter} = 2 \times (\text{Length} + \text{Breadth})$
 - $\text{Diagonal} = \sqrt{(\text{Length}^2 + \text{Breadth}^2)}$

Task 2: Standard Test Case Table:

Test Case	Input (Length m, Breadth m)	Expected Output	Actual Output	Result (Pass/Fail)
1	10, 4	Area = 400000 cm ² Perimeter = 2800 cm Diagonal = 1077.03 cm	Area = 400000 cm ² Perimeter = 2800 cm Diagonal = 1077.03 cm	PASS
2	12, 4.8	Area = 576000 cm ² Perimeter = 3360 cm Diagonal = 1292.44 cm	Area = 576000 cm ² Perimeter = 3360 cm Diagonal = 1292.44 cm	PASS
3	6, 6	Area = 360000 cm ² Perimeter = 2400 cm Diagonal = 848.53 cm	Area = 360000 cm ² Perimeter = 2400 cm Diagonal = 848.53 cm	PASS
4	5, 8.2	Area = 410000 cm ² Perimeter = 2640 cm Diagonal = 960.42 cm	Area = 410000 cm ² Perimeter = 2640 cmDiagonal = 960.42 cm	PASS
5	0, 7	Error: Length and Breadth must be greater than zero.	Error: Length and Breadth must be greater than zero.	PASS

Task 3: The Algorithm:

Step 1: Start the program.

Step 2: Declare variables for:

- Length (meters)
- Breadth (meters)
- Length (centimetres)
- Breadth (centimetres)
- Area
- Perimeter
- Diagonal

Step 3: Read the length and breadth in meters.

Step 4: Check whether the entered values are greater than zero.

- If either value is less than or equal to zero, display an error message and terminate the program.

Step 5: Convert length and breadth from meters to centimetres.

Step 6: Calculate:

- Area = Length × Breadth
- Perimeter = 2 × (Length + Breadth)
- Diagonal = $\sqrt{\text{Length}^2 + \text{Breadth}^2}$

Step 7: Display the calculated area, perimeter, and diagonal.

Step 8: Stop the program.

Evaluation of the Algorithm

- The algorithm is simple and easy to understand.
- It correctly converts the units before performing calculations.
- It validates the input to prevent invalid dimensions.
- It produces accurate values for all valid rectangles.

Task 4: C program:

The c program for the above scenario will be:-

```
#include <stdio.h>
#include <math.h>
int main()
{
    float length_m, breadth_m;
    float length_cm, breadth_cm;
    float area, perimeter, diagonal;
    printf("Enter Length (in meters): ");
    scanf("%f", &length_m);
    printf("Enter Breadth (in meters): ");
    scanf("%f", &breadth_m);
    if(length_m <= 0 || breadth_m <= 0)
    {
        printf("\nError: Length and Breadth must be greater than zero.\n");
        return 1;
    }

    length_cm = length_m * 100;
    breadth_cm = breadth_m * 100;
    area = length_cm * breadth_cm;
    perimeter = 2 * (length_cm + breadth_cm);
    diagonal = sqrt(length_cm * length_cm + breadth_cm * breadth_cm);
    printf("\nRectangle Details\n");
    printf("-----\n");
    printf("Length      : %.2lf cm\n", length_cm);
    printf("Breadth     : %.2lf cm\n", breadth_cm);
    printf("Area        : %.2lf sq. cm\n", area);
    printf("Perimeter   : %.2lf cm\n", perimeter);
    printf("Diagonal    : %.2lf cm\n", diagonal);
    return 0;
}
```

Task 5: Test cases output:

Test case1:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter Length of the rectangle (in meters): 10
Enter breadth of the rectangle (in meters): 4

=====Rectangle Details=====
Length      : 1000.00 cm
Breadth     : 400.00 cm
Area        : 400000.00 sq.cm
Perimeter   : 2800.00 cm
Diagonal    : 1077.03 cm
```

Test case2:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter Length of the rectangle (in meters): 12
Enter breadth of the rectangle (in meters): 4.8

=====Rectangle Details=====
Length      : 1200.00 cm
Breadth     : 480.00 cm
Area        : 576000.06 sq.cm
Perimeter   : 3360.00 cm
Diagonal    : 1292.44 cm
```

Test case3:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter Length of the rectangle (in meters): 6
Enter breadth of the rectangle (in meters): 6

=====Rectangle Details=====
Length      : 600.00 cm
Breadth     : 600.00 cm
Area        : 360000.00 sq.cm
Perimeter   : 2400.00 cm
Diagonal    : 848.53 cm
```

Test case4:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter Length of the rectangle (in meters): 5
Enter breadth of the rectangle (in meters): 8.2

=====Rectangle Details=====
Length      : 500.00 cm
Breadth     : 820.00 cm
Area        : 410000.00 sq.cm
Perimeter   : 2640.00 cm
Diagonal    : 960.42 cm
```

Test case5:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter Length of the rectangle (in meters): 0
Enter breadth of the rectangle (in meters): 7

Error: Length and Breadth must be greater than zero.
```

Question 2:-

Write a C program to find the perimeter and area of a regular pentagon when its side and apothem are given.

Task1: Understand the Problem Statement:

- **Input**
 - Side of the regular pentagon
 - Apothem of the regular pentagon
- **Output**
 - Perimeter of the pentagon
 - Area of the pentagon
- **Formulae**
 - $\text{Perimeter} = 5 \times \text{Side}$
 - $\text{Area} = (\text{Perimeter} \times \text{Apothem}) / 2$

Task 2: Standard Test Case Table:

Test Case	Input (Side, Apothem)	Expected Output	Actual Output	Result (Pass/Fail)
1	6, 4.1	Perimeter = 30.00 units Area = 61.50 sq. units	Perimeter = 30.00 units Area = 61.50 sq. units	PASS
2	8, 5.5	Perimeter = 40.00 units Area = 110.00 sq. units	Perimeter = 40.00 units Area = 110.00 sq. units	PASS
3	10, 6.9	Perimeter = 50.00 units Area = 172.50 sq. units	Perimeter = 50.00 units Area = 172.50 sq. units	PASS
4	12, 8.3	Perimeter = 60.00 units Area = 249.00 sq. units	Perimeter = 60.00 units Area = 249.00 sq. units	PASS
5	0, 5	Error: Side and Apothem must be greater than zero.	Error: Side and Apothem must be greater than zero.	PASS

Task 3: The Algorithm:

Step 1: Start the program.

Step 2: Declare variables for side, apothem, perimeter, and area.

Step 3: Read the side and apothem from the user.

Step 4: Check whether the side and apothem are greater than zero.

- If either value is less than or equal to zero, display an error message and terminate the program.

Step 5: Calculate the perimeter using:

$\text{Perimeter} = 5 \times \text{Side}$

Step 6: Calculate the area using:

$\text{Area} = (\text{Perimeter} \times \text{Apothem}) / 2$

Step 7: Display the perimeter and area.

Step 8: Stop the program.

➤ Evaluation of the Algorithm

- The algorithm is simple and easy to understand.
- It uses the standard mathematical formulae for a regular pentagon.
- Input validation prevents invalid calculations.
- It gives accurate results for all valid inputs.

Task 4: C program:

```
#include <stdio.h>
int main()
{
    float side, apothem;
    float perimeter, area;
    printf("Enter the side of the regular pentagon: ");
    scanf("%f", &side);
    printf("Enter the apothem of the regular pentagon: ");
    scanf("%f", &apothem);
    if (side <= 0 || apothem <= 0)
    {
        printf("\nError: Side and Apothem must be greater than zero.\n");
        return 1;
    }

    perimeter = 5 * side;
    area = (perimeter * apothem) / 2;
    printf("\n===== Regular Pentagon Details =====\n");
    printf("Side : %.2f units\n", side);
    printf("Apothem : %.2f units\n", apothem);
    printf("Perimeter : %.2f units\n", perimeter);
    printf("Area : %.2f sq. units\n", area);
    return 0;
}
```

Task 5: Test cases output:

Test case1:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter the side of the regular pentagon: 6
Enter the apothem of the regular pentagon: 4.1

===== Regular Pentagon Details =====
Side      : 6.00 units
Apothem   : 4.10 units
Perimeter : 30.00 units
Area      : 61.50 sq. units
```

Test case2:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter the side of the regular pentagon: 8
Enter the apothem of the regular pentagon: 5.5

===== Regular Pentagon Details =====
Side      : 8.00 units
Apothem   : 5.50 units
Perimeter : 40.00 units
Area      : 110.00 sq. units
```

Test case3:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter the side of the regular pentagon: 10
Enter the apothem of the regular pentagon: 6.9

===== Regular Pentagon Details =====
Side      : 10.00 units
Apothem   : 6.90 units
Perimeter : 50.00 units
Area      : 172.50 sq. units
```

Test case4:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter the side of the regular pentagon: 12
Enter the apothem of the regular pentagon: 8.3

===== Regular Pentagon Details =====
Side      : 12.00 units
Apothem   : 8.30 units
Perimeter : 60.00 units
Area      : 249.00 sq. units
```

Test case5:

```
(kali㉿kali)-[~/HDOC6.c]
$ ./a.out
Enter the side of the regular pentagon: 0
Enter the apothem of the regular pentagon: 5

Error: Side and Apothem must be greater than zero.

(kali㉿kali)-[~/HDOC6.c]
$
```